

Bike Sharing Dataset

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2.a Variable Types

Identify which predictors are quantitative and which are qualitative.

```
library(tidyverse)
library(kableExtra)

day_data <- read.csv(
  "data/bike+sharing+dataset/day.csv",
  sep = ",",
)

hour_data <- read.csv(
  "data/bike+sharing+dataset/hour.csv",
  sep = ",",
)

# Quantitative predictor variables
day_quantitative_predictors <- c(
  "temp", "atemp", "hum", "windspeed"
)

hour_quantitative_predictors <- c(
  "temp", "atemp", "hum", "windspeed"
)

# Qualitative predictor variables
day_qualitative_predictors <- c(
  "season", "yr", "mnth", "holiday",
```

```

    "weekday", "workingday", "weathersit"
  )

hour_qualitative_predictors <- c(
  "season", "yr", "mnth", "hr", "holiday",
  "weekday", "workingday", "weathersit"
)

# Datetime variable
day_datetime <- c("dteday")
hour_datetime <- c("dteday")

# Index identifier
day_index <- c("instant")
hour_index <- c("instant")

# Response/count variables
day_response <- c("casual", "registered", "cnt")
hour_response <- c("casual", "registered", "cnt")

# First six observations of the day dataset
table_1 <- knitr::kable(
  head(day_data[, 1:8]),
  caption = "First six observations of the day dataset."
) %>%
  kableExtra::kable_styling(
    latex_options = "HOLD_position"
  )

table_2 <- knitr::kable(
  head(day_data[, 9:16]),
  caption = "First six observations of the day dataset."
) %>%
  kableExtra::kable_styling(
    latex_options = "HOLD_position"
  )

# First six observations of the hour dataset
table_3 <- knitr::kable(
  head(hour_data[, 1:9]),

```

```

    caption = "First six observations of the hour dataset."
  ) %>%
    kableExtra::kable_styling(
      latex_options = "HOLD_position"
    )

table_4 <- knitr::kable(
  head(hour_data[, 10:17]),
  caption = "First six observations of the hour dataset."
) %>%
  kableExtra::kable_styling(
    latex_options = "HOLD_position"
  )

# Variable types and roles
variable_types <- data.frame(
  Dataset = c(
    "Day", "Day", "Day", "Day", "Day",
    "Hour", "Hour", "Hour", "Hour", "Hour"
  ),
  Type = c(
    "Quantitative Predictor",
    "Qualitative Predictor",
    "Datetime",
    "Index Identifier",
    "Response/Count",
    "Quantitative Predictor",
    "Qualitative Predictor",
    "Datetime",
    "Index Identifier",
    "Response/Count"
  ),
  Variables = c(
    paste(day_quantitative_predictors, collapse = ", "),
    paste(day_qualitative_predictors, collapse = ", "),
    paste(day_datetime, collapse = ", "),
    paste(day_index, collapse = ", "),
    paste(day_response, collapse = ", "),
    paste(hour_quantitative_predictors, collapse = ", "),
    paste(hour_qualitative_predictors, collapse = ", "),
    paste(hour_datetime, collapse = ", ")
  )

```

```

    paste(hour_index, collapse = ", "),
    paste(hour_response, collapse = ", ")
  )
)

table_5 <- knitr::kable(
  variable_types,
  caption = "Variable Types and Roles in the Day and Hour Datasets.",
  col.names = c("Dataset", "Type", "Variables")
) %>%
  kableExtra::kable_styling(
    latex_options = "HOLD_position"
  )

```

Table 1: First six observations of the day dataset.

instant	dteday	season	yr	mnth	holiday	weekday	workingday
1	2011-01-01	1	0	1	0	6	0
2	2011-01-02	1	0	1	0	0	0
3	2011-01-03	1	0	1	0	1	1
4	2011-01-04	1	0	1	0	2	1
5	2011-01-05	1	0	1	0	3	1
6	2011-01-06	1	0	1	0	4	1

Table 2: First six observations of the day dataset.

weathersit	temp	atemp	hum	windspeed	casual	registered	cnt
2	0.344167	0.363625	0.805833	0.1604460	331	654	985
2	0.363478	0.353739	0.696087	0.2485390	131	670	801
1	0.196364	0.189405	0.437273	0.2483090	120	1229	1349
1	0.200000	0.212122	0.590435	0.1602960	108	1454	1562
1	0.226957	0.229270	0.436957	0.1869000	82	1518	1600
1	0.204348	0.233209	0.518261	0.0895652	88	1518	1606

Table 3: First six observations of the hour dataset.

instant	dteday	season	yr	mnth	hr	holiday	weekday	workingday
1	2011-01-01	1	0	1	0	0	6	0

2	2011-01-01	1	0	1	1	0	6	0
3	2011-01-01	1	0	1	2	0	6	0
4	2011-01-01	1	0	1	3	0	6	0
5	2011-01-01	1	0	1	4	0	6	0
6	2011-01-01	1	0	1	5	0	6	0

Table 4: First six observations of the hour dataset.

weathersit	temp	atemp	hum	windspeed	casual	registered	cnt
1	0.24	0.2879	0.81	0.0000	3	13	16
1	0.22	0.2727	0.80	0.0000	8	32	40
1	0.22	0.2727	0.80	0.0000	5	27	32
1	0.24	0.2879	0.75	0.0000	3	10	13
1	0.24	0.2879	0.75	0.0000	0	1	1
2	0.24	0.2576	0.75	0.0896	0	1	1

Table 5: Variable Types and Roles in the Day and Hour Datasets.

Dataset	Type	Variables
Day	Quantitative Predictor	temp, atemp, hum, windspeed
Day	Qualitative Predictor	season, yr, mnth, holiday, weekday, workingday, weathersit
Day	Datetime	dteday
Day	Index Identifier	instant
Day	Response/Count	casual, registered, cnt
Hour	Quantitative Predictor	temp, atemp, hum, windspeed
Hour	Qualitative Predictor	season, yr, mnth, hr, holiday, weekday, workingday, weathersit
Hour	Datetime	dteday
Hour	Index Identifier	instant
Hour	Response/Count	casual, registered, cnt

Predictor variables are used to explain or predict a response variable. Quantitative predictors have numerical values that represent measurable quantities, while qualitative predictors represent categories. The response variables are the outcomes being predicted. The index identifier is used to identify observations. The datetime variable is its own variable type, rather than being a qualitative predictor, because dates have meaningful chronological order rather than being labels for unrelated categories.

We can automatically check the R storage type of each variable. However, the storage type does not always determine the statistical type. For example, a qualitative variable can still

have a numeric data type. Because of this, there is no reliable way to assign all variable types automatically based on the data type, so the variable types are assigned manually.

2.b Summary Statistics

For each quantitative predictor, calculate and report:

- Range
- Mean
- Standard deviation

Then determine which season has the highest average bike rental count.

```
library(kableExtra)

# Summary statistics for quantitative predictors
day_summary <- day_data %>%
  summarise(
    across(
      all_of(day_quantitative_predictors),
      list(
        Range = ~ max(.x) - min(.x),
        Mean = ~ mean(.x),
        SD = ~ sd(.x)
      )
    )
  ) %>%
  pivot_longer(
    everything(),
    names_to = c("Variable", "Statistic"),
    names_sep = "_"
  ) %>%
  pivot_wider(
    names_from = Statistic,
    values_from = value
  ) %>%
  mutate(
    Mean = round(Mean, 2),
    SD = round(SD, 2)
  )

table_summary <- knitr::kable(
```

```

day_summary,
caption = "Summary Statistics for Quantitative Predictors.",
booktabs = TRUE
) %>%
kableExtra::kable_styling(
  latex_options = "hold_position"
)

# Average bike rental count by season
season_summary <- day_data %>%
  group_by(season) %>%
  summarise(
    average_rentals = mean(cnt)
  ) %>%
  arrange(desc(average_rentals)) %>%
  mutate(
    average_rentals = round(average_rentals, 2)
  )

table_season <- knitr::kable(
  season_summary,
  col.names = c("Season", "Average Rentals"),
  caption = "Average Bike Rentals by Season.",
  booktabs = TRUE
) %>%
kableExtra::kable_styling(
  latex_options = "hold_position"
)

# Identify season with highest average rentals
highest_season <- season_summary %>%
  slice_max(average_rentals, n = 1)

```

Table 6: Summary Statistics for Quantitative Predictors.

Variable	Range	Mean	SD
temp	0.8025366	0.50	0.18
atemp	0.7618264	0.47	0.16
hum	0.9725000	0.63	0.14
windspeed	0.4850713	0.19	0.08

Table 7: Average Bike Rentals by Season.

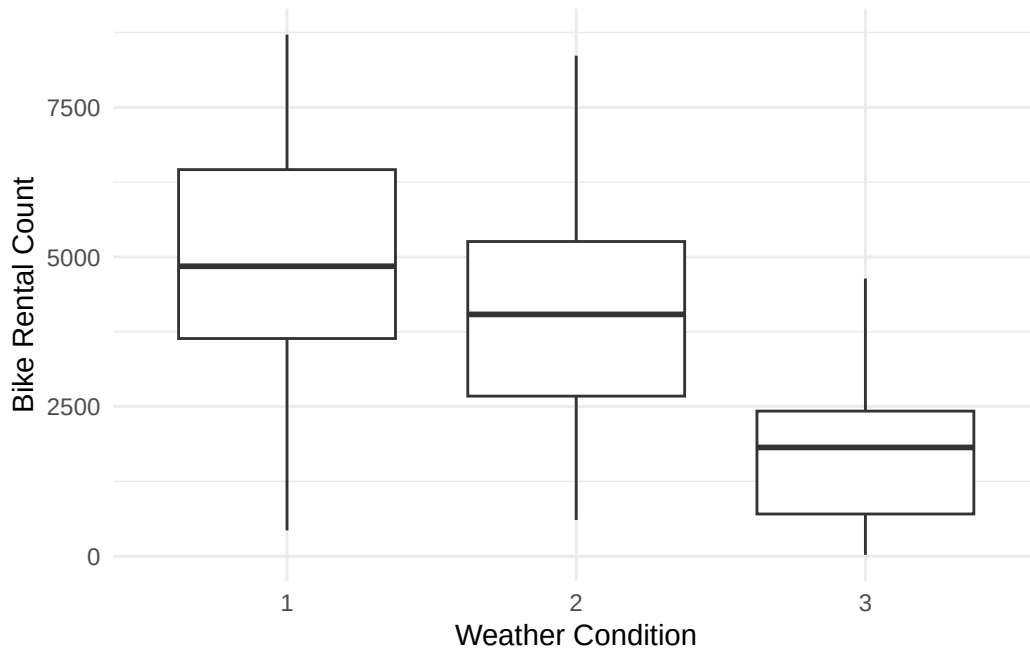
Season	Average Rentals
3	5644.30
2	4992.33
4	4728.16
1	2604.13

Season with the highest average bike rental count: 3
Average rental count: 5644.3

2.c Rentals by Weather Condition

Create boxplots showing bike rental count for each weather condition.

```
# Boxplots of bike rental count by weather condition
ggplot(day_data, aes(x = factor(weathersit), y = cnt)) +
  geom_boxplot() +
  labs(
    x = "Weather Condition",
    y = "Bike Rental Count"
  ) +
  theme_minimal()
```



```
# Median bike rental count by weather condition
weather_summary <- day_data %>%
  group_by(weathersit) %>%
  summarise(
    median_rentals = median(cnt)
  ) %>%
  arrange(desc(median_rentals)) %>%
  mutate(
    median_rentals = round(median_rentals, 2)
  )

table_weather <- knitr::kable(
  weather_summary,
  col.names = c("Weather Condition", "Median Rentals"),
  caption = "Median Bike Rentals by Weather Condition.",
  booktabs = TRUE
) %>%
  kableExtra::kable_styling(
    latex_options = "hold_position"
  )
```

Table 8: Median Bike Rentals by Weather Condition.

Weather Condition	Median Rentals
1	4844
2	4040
3	1817

Weather condition with the highest median bike rental count: 1

The median bike rental count is highest under weather condition 1, with a median of 4,844 rentals. Weather condition 2 has a lower median of 4,040, while condition 3 has the lowest at 1,817. Overall, bike rentals tend to be higher under weather condition 1.

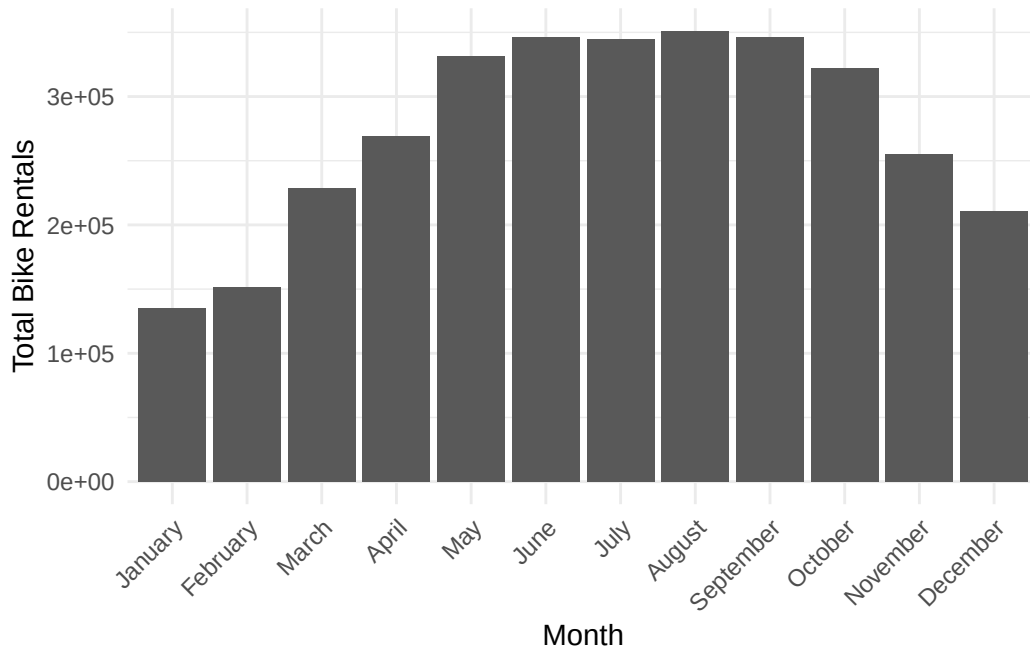
2.d Monthly Bike Rentals

Create a bar plot showing the total number of bike rentals for each month of the year.

```
# Calculate total bike rentals for each month
monthly_rentals <- day_data %>%
  mutate(
    month = factor(
      format(as.Date(dteday), "%B"),
      levels = month.name,
      ordered = TRUE
    )
  ) %>%
  group_by(month) %>%
  summarise(
    total_rentals = sum(cnt)
  )

# Bar plot of monthly rental totals
ggplot(monthly_rentals, aes(x = month, y = total_rentals)) +
  geom_col() +
  labs(
    x = "Month",
    y = "Total Bike Rentals"
  ) +
  theme_minimal() +
```

```
theme(
  axis.text.x = element_text(angle = 45, hjust = 1)
)
```



```
# Format monthly rental totals as a table
table_monthly <- knitr::kable(
  monthly_rentals,
  col.names = c("Month", "Total Rentals"),
  caption = "Total Bike Rentals by Month.",
  booktabs = TRUE
) %>%
  kableExtra::kable_styling(
    latex_options = c("hold_position")
  )

# Identify month with highest total rentals
highest_month <- monthly_rentals %>%
  slice_max(total_rentals, n = 1)
```

Table 9: Total Bike Rentals by Month.

Month	Total Rentals
January	134933
February	151352
March	228920
April	269094
May	331686
June	346342
July	344948
August	351194
September	345991
October	322352
November	254831
December	211036

Month with the highest total number of rentals: August
Total rentals: 351194

I extracted the month from the `dteday` variable, grouped the data by month, and summed the `cnt` variable within each month to calculate the total number of bike rentals. The month with the highest number of rentals is **August**, with **351194** total rentals.

2.e Relationships Among Variables

Using the dataset, investigate the relationships among the predictors. Focus the correlation matrix on **quantitative predictors**: `temp`, `atemp`, `hum`, and `windspeed`. The rental-count variables (`casual`, `registered`, `cnt`) are response variables rather than predictors. Avoid including variables such as `season`, `weekday`, `weathersit` because their categories are represented by numerical codes.

```
# Quantitative predictors
quantitative_predictors <- c(
  "temp",
  "atemp",
  "hum",
  "windspeed"
)

# Correlation matrix
correlation_matrix <- day_data %>%
```

```

select(all_of(quantitative_predictors)) %>%
cor() %>%
round(3)

table_correlation <- knitr::kable(
  correlation_matrix,
  caption = "Correlation Matrix of Quantitative Predictors.",
  booktabs = TRUE
) %>%
  kableExtra::kable_styling(latex_options = "hold_position")

table_correlation

```

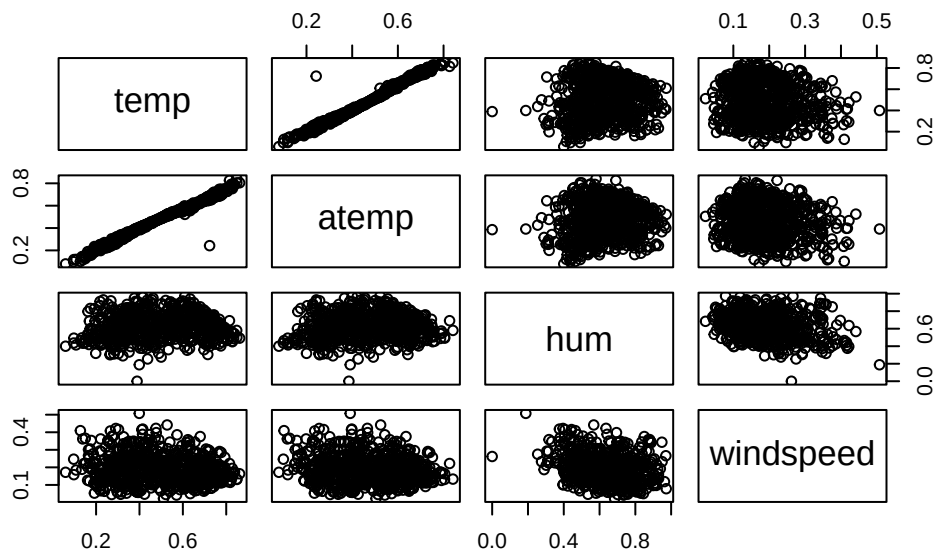
Table 10: Correlation Matrix of Quantitative Predictors.

	temp	atemp	hum	windspeed
temp	1.000	0.992	0.127	-0.158
atemp	0.992	1.000	0.140	-0.184
hum	0.127	0.140	1.000	-0.248
windspeed	-0.158	-0.184	-0.248	1.000

```

# Scatterplot matrix
day_data %>%
  select(all_of(quantitative_predictors)) %>%
  pairs()

```



The strongest relationship is between **temp** and **atemp** with a correlation of 0.992, indicating a strong positive linear relationship. This is also seen in the scatterplot. **temp** and **atemp** are highly redundant.

2.f Predicting Bike Rental Count

We can identify potential predictors and examine redundancy based on the relationships found in 2.e. Since **temp** and **atemp** are highly correlated, using both as predictors could introduce redundancy. For a prediction model, I would use one of these variables along with hum and windspeed. Categorical variables such as season, yr, and weathersit could also be used.

The correlation analysis alone cannot determine which variables are most useful for prediction, because it only examines linear relationships among quantitative predictors. It does not measure the relationships between the quantitative predictors and the response variable. Furthermore, it does not capture nonlinear relationships or relationships involving categorical variables.